

Academic References and Further Reading

EE5203 Microprocessors — where the full story is

The lectures in this course introduce each concept **conceptually**: enough mechanism to configure, use, and debug it, without the full architectural detail. This page maps every simplified concept to the authoritative document where the complete story lives, so that a curious student — or a future project — can go one level deeper with a known-good source instead of a random blog.

Each lecture gets its own section. Entries are added as lectures are revised.

If a link stops working

Links below point at the current official locations (verified 2026-07-17). If one rots:

- **ST documents** (RM..., PM..., UM..., MB... numbers): search the document number at [st.com](https://www.st.com); the number uniquely identifies the document and survives site reorganizations.
- **Arm documents** (DDI... numbers): search at developer.arm.com.
- **Books**: available through the university library or the cited publisher; Lee & Seshia is free and legal at leeseshia.org.

Primary documents used throughout the course

Short name	Full citation	What it is for
RM0368	STMicroelectronics, <i>RM0368:</i> <i>STM32F401xB/C and</i> <i>STM32F401xD/E</i> <i>advanced Arm®-based</i> <i>32-bit MCUs —</i> <i>Reference Manual</i> (PDF)	The register-level truth for every peripheral on our chip: GPIO, SYSCFG, EXTI, timers, ADC, I2C, SPI, DMA

Short name	Full citation	What it is for
PM0214	STMicroelectronics, <i>PM0214: STM32 Cortex®-M4 MCUs and MPUs programming manual</i> (PDF)	The processor core seen from software: instruction set, exception model, NVIC, SysTick, core registers
UM1724	STMicroelectronics, <i>UM1724: STM32 Nucleo-64 boards (MB1136) — User Manual</i> (PDF)	The board itself: B1 and LD2 circuits, jumpers, ST-LINK, power, board revision differences
MB1136 schematic	STMicroelectronics, <i>MB1136 schematic pack</i> , rev C-04 (PDF)	The actual circuit diagram of the Nucleo-F401RE — the version used in lecture
UM1725	STMicroelectronics, <i>UM1725: Description of STM32F4 HAL and low-layer drivers</i> (PDF)	Signature and behavior of every <code>HAL_...</code> function the labs use
Yiu 2014	J. Yiu, <i>The Definitive Guide to ARM® Cortex®-M3 and Cortex®-M4 Processors</i> , 3rd ed., Newnes, 2014. ISBN 978-0-12-408082-9	The standard book-length treatment of the core: pipeline, exceptions, NVIC, low-power, debug
DDI 0403	Arm Ltd., <i>ARMv7-M Architecture Reference Manual</i> (Arm DDI 0403)	The formal architecture specification — for when even Yiu is not precise enough
Lee & Seshia	E. A. Lee and S. A. Seshia, <i>Introduction to Embedded Systems: A Cyber-Physical Systems Approach</i> , 2nd ed., MIT Press, 2017 — free PDF	The academic textbook view: interrupts, concurrency, real-time reasoning
White 2024	E. White, <i>Making Embedded Systems</i> , 2nd ed., O'Reilly, 2024	Practical engineering judgment: interrupt hygiene, debouncing, system design patterns
C11	ISO/IEC 9899:2011, <i>Programming languages — C</i> (link is N1570, the final public draft, PDF)	The C language standard; final authority on <code>volatile</code> , integer types, undefined behavior

L01 — Embedded Systems, the C Toolchain, and Debugging

Concept as taught	Where the full story is
What an embedded system is — dedicated function + constraints The Nucleo-F401RE board — ST-LINK, LD2, B1, headers Processor / MCU / system model The toolchain — compile, link, flash, debug First C statements — <code>main()</code> , declarations, <code>if</code> , <code>while</code> (1)	Lee & Seshia , ch. 1 <i>Introduction</i> (free PDF); White 2024 , ch. 1 UM1724 Rev 17 , §6 <i>Hardware layout</i> — p. 13; MB1136 C-04 schematic PM0214 Rev 10 , §2.1 <i>Programmers model</i> — p. 17; Yiu 2014, ch. 1–2 Yiu 2014, ch. 2 (development flow and what CMSIS standardizes); the VS Code build terminal of any generated project shows each stage’s real output C11 (N1570) , §6.8.4 <i>Selection statements</i> — p. 166; program startup is §5.1.2.2.1

L02 — Values, Decisions, Loops, and Arrays

Concept as taught	Where the full story is
if / else if / else	C11 (N1570) , §6.8.4 <i>Selection statements</i> — p. 166
while and for loops	C11 (N1570) , §6.8.5 <i>Iteration statements</i> — p. 168
Arrays and indexing from zero	C11 (N1570) , §6.5.2.1 <i>Array subscripting</i> — p. 98 — out-of-range access is undefined behavior, which is why <code>samples[8]</code> is not “just the next value”
uint16_t and friends	C11 (N1570) , §7.20 <i>Integer types</i> <code><stdint.h></code> — p. 307
Integer division — <code>a / b</code> truncates	C11 §6.5.5 (multiplicative operators): division truncates toward zero

L03 — Functions, Arrays, Strings, and Multiple C Files

Concept as taught	Where the full story is
Declarations, definitions, calls	C11 (N1570) , §6.9.1 <i>Function definitions</i> — p. 174
Pass by value — parameters are copies	C11 (N1570) , §6.5.2.2 <i>Function calls</i> — p. 99

Concept as taught	Where the full story is
Headers, multiple files, the linker	C11 (N1570), §6.2.2 <i>Linkages of identifiers</i> — p. 54; <code>#include</code> / <code>#define</code> are §6.10 <i>Preprocessing directives</i> — p. 184
Why array parameters need a count — arrays decay to pointers C strings and <code>\0</code>; <code>strcmp</code>	C11 §6.7.6.3¶7; the L04 pointer material makes this concrete C11 (N1570), §7.24 <i>String handling</i> <code><string.h></code> — p. 380

L04 — Pointers, Structures, Bits, Registers, and GPIO

The lecture builds `GPIOA->BSRR = (1U << 5);` from its parts: addresses, pointers, structures, masks, and the GPIO registers.

Concept as taught	Where the full story is
Addresses and memory — “memory is a numbered collection of storage locations” Pointers: <code>&</code> and <code>*</code>	PM0214 Rev 10, §2.2 <i>Memory model</i> — p. 28: the real Cortex-M4 memory map, including where peripherals live C11 (N1570), §6.5.3.2 <i>Address and indirection operators</i> — p. 106
Structures and <code>.</code> / <code>-></code>	C11 (N1570), §6.7.2.1 <i>Structure and union specifiers</i> — p. 130
Bitwise operators, masks, the U suffix	C11 (N1570), §6.5.7 <i>Bitwise shift operators</i> — p. 112, then §6.5.10–6.5.12 for <code>&</code> , <code>^</code> , <code> </code> ; §6.5.7 also explains why shifting signed values can be undefined behavior — the reason the U suffix matters
CMSIS register names — “the device header supplies the definition”	The actual <code>stm32f401xe.h</code> device header (<code>GPIO_TypeDef</code> , <code>GPIOA_BASE</code>); Yiu 2014, ch. 2 on what CMSIS standardizes
volatile register members	C11 (N1570), §6.7.3 — p. 140; M. Barr, <i>How to Use C’s volatile Keyword</i>
Peripheral clock gating — “enable the clock first”	RM0368 Rev 6, <code>RCC_AHB1ENR</code> — p. 118; the RCC chapter (§6) explains <i>why</i> clocks are gated: power
GPIO block, MODER, BSRR	RM0368 Rev 6, §8 <i>GPIO</i> — p. 146; registers: <code>GPIOx_MODER</code> — p. 158, <code>GPIOx_BSRR</code> — p. 161 — including what the lecture skips: <code>OTYPER</code> , <code>OSPEEDR</code> , <code>PUPDR</code>

L05 — GPIO and Hardware Abstraction

The lecture connects every HAL name to the register behavior taught in L04: same hardware, second interface.

Concept as taught	Where the full story is
The HAL GPIO API — GPIO_InitTypeDef, HAL_GPIO_Init, WritePin / TogglePin / ReadPin What the HAL actually executes	UM1725 Rev 8, GPIO functions — p. 447 The source of stm32f4xx_hal_gpio.c — HAL_GPIO_WritePin is a single BSRR write RM0368 Rev 6, §8 GPIO — p. 146 ; registers: GPIOx_MODER — p. 158 , GPIOx_PUPDR — p. 159 , GPIOx_BSRR — p. 161
HAL names → register bits — GPIO_MODE_OUTPUT_PP, GPIO_NOPULL, speed	RM0368 §8.3 (I/O port control) ; the electronics context in any circuits text — the lecture’s claim is only that an undriven CMOS input is undefined MB1136 schematic rev C-04 ; UM1724 Rev 17, push-buttons — p. 24 ; confirmed against the ST Nucleo BSP source
Floating inputs, pull-up / pull-down	White 2024 , ch. 2 (creating system diagrams / layering); Yiu 2014, ch. 2 (what CMSIS standardizes and why)
B1 circuit — external pull-up, active-low	Forward pointer: the L06 row <i>Polling vs. interrupts</i> above
Why abstraction layers exist — and what they cost	
Polling and its limits — “the loop asks again and again”	

L06 — GPIO Interrupts

The lecture walks one button press through five plain-language stages (PRESS → NOTICE → ASK → LOOK UP → RUN YOUR CODE). Each simplification below names the document that removes it.

Page-anchored links ([#page=N](#)) below open the PDF at the cited page in the browser viewer. Page numbers were verified against the document revisions named in each entry (2026-07-17); if ST publishes a new revision, the section numbers in the text remain the reliable pointer.

Concept as taught	Where the full story is
Polling vs. interrupts — “polling asks again and again”	Lee & Seshia , ch. 10 <i>Input and Output</i> (interrupt mechanisms, latency, and when polling is actually the better design); Yiu 2014, §4.5
What really happens between ASK and RUN YOUR CODE — register stacking, vector fetch, exception entry/return (we skipped this entirely)	PM0214 Rev 10 , §2.3 <i>Exception model</i> — p. 37; Yiu 2014, ch. 8 (including the 12-cycle entry sequence and tail-chaining); DDI 0403 , §B1.5 for the formal definition
EXTI — “a set of hardware edge detectors”	RM0368 Rev 6 , §10 <i>EXTI</i> — p. 205: block diagram, IMR/EMR/RTSR/FTSR/SWIER/PR registers, the event-vs-interrupt distinction we did not cover
SYSCFG routing — “a four-bit field selects the port”	RM0368 Rev 6 , <i>SYSCFG_EXTICR4</i> — p. 143
Pending bit and write-1-to-clear	RM0368 Rev 6 , <i>EXTI_PR</i> — p. 211; the same <code>rc_w1</code> convention appears in many STM32 peripherals
NVIC — “enable, priority, deliver”	PM0214 Rev 10 , §4.2 <i>NVIC</i> — p. 208; Yiu 2014, ch. 7 — including what we postponed: priority levels, preemption, sub-priority, nesting
Vector table and the startup file	PM0214 Rev 10 , §2.3.4 <i>Vector table</i> — p. 40; Yiu 2014, §4.4 and §8.5; the generated <code>startup_stm32f401retx.s</code> in every CubeMX-generated project is the concrete instance
EXTI15_10_IRQHandler / HAL helper / callback chain	UM1725 Rev 8 , <i>GPIO functions</i> — p. 447 (<code>HAL_GPIO_EXTI_IRQHandler</code> , <code>HAL_GPIO_EXTI_Callback</code>); the ~10-line source of the helper in <code>stm32f4xx_hal_gpio.c</code> is worth reading in the lab
volatile — “read the stored value each time”	C11 (N1570) , §6.7.3 — p. 140; M. Barr, <i>How to Use C's volatile Keyword</i> , Barr Group. Note what the lecture only hints at: <code>volatile</code> does not make read-modify-write atomic — see Lee & Seshia , ch. 11 on shared-variable concurrency

Concept as taught	Where the full story is
ISR <code>main()</code> shared-variable pattern (<code>button_event</code>)	Lee & Seshia , ch. 11 (race conditions, atomicity); White 2024 , interrupt chapter (keep-ISRs-short discipline and its reasons)
Switch bounce and the 80 ms guard	J. Ganssle, <i>A Guide to Debouncing</i> , 2nd ed., The Ganssle Group, 2008: measured bounce data for real switches (the “settles within ~10 ms” claim comes from here), plus hardware and software debounce alternatives
B1 circuit — external pull-up, active-low	MB1136 schematic rev C-04 ; UM1724 Rev 17, push-buttons — p. 24 — check the board revision : later MB1136 revisions changed the B1 circuit, which affects the pull configuration the lab assumes

L07 — Timers

The lecture builds one mental model — clock → PSC → CNT → ARR → update event — and consumes it by polling, by interrupt, and as a 1 ms tick.

Concept as taught	Where the full story is
The time base — prescaler, counter, auto-reload, update event	RM0368 Rev 6, §13 <i>TIM2 to TIM5</i> — p. 316; registers (§13.4 — p. 353)
Where the 16 MHz comes from — clock tree, APB timer-clock ×2 rule	RM0368 Rev 6, §6.2 <i>Clocks</i> — p. 94
HAL TIM base API — handle, <code>Base_Init</code> , <code>Base_Start_IT</code> , callback	UM1725 Rev 8, TIM functions — p. 1065
The interrupt path — reused from L06	The L06 rows above; the only new element is the request name <code>TIM2_IRQn</code>
Tick-based scheduling and why blocking delays hurt	Lee & Seshia , ch. 10–11; White 2024 , timer and main-loop chapters
SysTick — the Cortex-M core timer behind <code>HAL_Delay</code> (not covered)	PM0214 Rev 10, §4.5 <i>SysTick timer</i> — p. 246

L08 — Pulse-Width Modulation

The lecture adds the capture/compare channel to L07’s time base and uses it in the compare direction: ARR owns the frequency, CCR owns the duty.

Concept as taught	Where the full story is
PWM mode — pin high while CNT < CCR	RM0368 Rev 6, §13.3.9 <i>PWM mode</i> — p. 336, including what the lecture skips: center-aligned mode, PWM mode 2
Channel registers	TIMx_CCMR1 — p. 362, TIMx_CCER — p. 366, TIMx_CCR1 — p. 367
Handing the pin to the timer — alternate function	RM0368 Rev 6, GPIOx_AFRL — p. 162; the pin-to-AF table is the STM32F401RE datasheet (DS10086), <i>Alternate function mapping</i> — search the DS number at st.com
HAL PWM API	UM1725 Rev 8, TIM functions — p. 1065
The servo pulse contract — 50 Hz frame, 1–2 ms pulse	Hobby-servo convention, not an ST document: any RC-servo datasheet (e.g., SG90, S3003) states the same 20 ms / 1–2 ms numbers
Why the average works — PWM into slow loads	White 2024 , output/actuator chapters; the RC-filter demo linked from the deck shows it interactively

L09 — Input Capture

The same channel, reversed: an edge on the pin captures the free-running counter into CCR — a hardware timestamp.

Concept as taught	Where the full story is
Input capture — edge copies CNT into CCR	RM0368 Rev 6, §13.3.5 <i>Input capture mode</i> — p. 333
Automated duty measurement	RM0368 Rev 6, §13.3.6 <i>PWM input mode</i> — p. 334 — the hardware version of the paired-channel trick, at the cost of the whole timer
Direct vs indirect inputs, the filter, overcapture	TIMx_CCMR1 — p. 362 (CCxS, ICF), TIMx_CCER — p. 366; overcapture flags in the TIMx_SR description
HAL capture API	UM1725 Rev 8, TIM functions — p. 1065; HAL_TIM_ReadCapturedValue — p. 1097

Concept as taught	Where the full story is
Ultrasonic ranging — echo width → distance	HC-SR04 datasheet (community-published; no ST equivalent): 10 μ s trigger, echo width, the /58 cm constant

L10 — Analog-to-Digital Conversion

The lecture builds counts volts fluency on one convention ($mV = raw \times 3300 / 4095$), one clock ($PCLK2 \div 2 = 8$ MHz), and one converter (the SAR core).

Concept as taught	Where the full story is
The ADC itself — 12-bit SAR, one converter on the F401	RM0368 Rev 6, §11 <i>ADC</i> — p. 213
The ADC clock — $PCLK2 \div 2 = 8$ MHz, 36 MHz ceiling	§11.3.2 <i>ADC clock</i> — p. 215
Channels — 16 external doors + the internal sensors	§11.3.3 <i>Channel selection</i> — p. 215; §11.9 <i>Temperature sensor</i> — p. 226
Single vs continuous conversion	§11.3.4–11.3.5 — p. 216 ; what the lecture skips: injected channels, the analog watchdog, external triggers (§11.6 — p. 222)
Sampling time and source impedance	§11.5 <i>Channel-wise programmable sampling time</i> — p. 222 ; the hard electrical limits (max source resistance vs sampling time) are in the STM32F401RE datasheet (DS10086), <i>ADC characteristics</i>
ADC registers	§11.12 — p. 229 ; ADC_SR — p. 229 , ADC_CR1 — p. 230 , ADC_CR2 — p. 232 , ADC_SMPR2 — p. 234 , ADC_SQR3 — p. 237 , ADC_DR — p. 239
The analog pin mode — MODER = 11	RM0368 Rev 6, GPIOx_MODER — p. 158
HAL ADC API	UM1725 Rev 8, ADC generic driver — p. 56 : HAL_ADC_Start — p. 65 , HAL_ADC_PollForConversion / HAL_ADC_Start_IT — p. 66 , HAL_ADC_GetValue / HAL_ADC_ConvCpltCallback — p. 68 , HAL_ADC_ConfigChannel — p. 69
Sampling theory and aliasing — stated, not proven	Lee & Seshia , sensors/actuators and sampling chapters; White 2024 , analog-input chapters

L11 — The Bridge to Assembly

Every listing in the deck is real output of `arm-none-eabi-gcc 14.3` — the toolchain inside STM32CubeCLT — compiled for Cortex-M4 at `-O0` and `-O2`.

Concept as taught	Where the full story is
The register file — R0–R12, SP, LR, PC, xPSR	PM0214 Rev 10, §2.1.3 <i>Core registers</i> — p. 18
Thumb-2 — 16- and 32-bit encodings, the <code>.w</code> suffix	PM0214 Rev 10, §3.1 <i>Instruction set summary</i> — p. 50; §3.3.8 <i>Instruction width selection</i> — p. 68
Memory access — LDR/STR, offsets, PUSH/POP	§3.4 — p. 69; §3.4.2 <i>immediate offset</i> — p. 71; §3.4.7 <i>PUSH and POP</i> — p. 78
Data processing — arithmetic, logic, shifts, CMP, MOV	§3.5 — p. 81: ADD/SUB — p. 83, AND/ORR/EOR/BIC — p. 85, shifts — p. 62, CMP — p. 88, MOV/MVN — p. 89
Branches and conditions — B/BL/BX, condition codes, IT	B, BL, BX, BLX — p. 142; §3.3.7 <i>Conditional execution</i> — p. 65; the <code>-O2</code> ITE surprise: IT — p. 145
Calling convention — args in r0–r3, result in r0, LR	The Arm AAPCS32 (<i>Procedure Call Standard for the Arm Architecture</i>) — search “AAPCS32” at developer.arm.com ; the lecture teaches only the subset visible in the debugger
Interrupt entry stacking — same stack, hardware-driven	the L06 rows above (exception model); the deck only points back to them
What the optimizer may do — folding, inlining, <code>volatile</code> limits	GCC manual, <i>Optimize Options</i> ; White 2024 , debugging/optimization chapters

Sections for other lectures are added as each lecture is revised to the 2026–2027 pattern.